

Name: _____

UNIT 3B: MENDELIAN GENETICS

Topic 3: Other Modes of Inheritance

Mode of Inheritance	Definition	Example
Incomplete Dominance		
Co-dominance		
Sexlinked traits		
Barr Bodies		
Multiple Alleles		

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PRACTICE PROBLEMS

1) In some varieties of flowering plants, petal colour is determined by two alleles which demonstrate incomplete dominance. One version of the allele, R , encodes red flowers, while another version of the allele, r , encodes white flowers.

a. Determine the genotype of each of the following plants:

- A plant with red flowers

- A plant with white flowers

- A plant with pink flowers

b. A plant with red flowers is crossed with a plant with pink flowers. What proportion of offspring will also have pink flowers?

c. Is it possible for the cross described in part b to produce offspring with white flowers? Explain.

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2) In some varieties of flowering plants, petal colour is determined by two alleles which demonstrate incomplete dominance. One version of the allele, R_1 , encodes red flowers, while another version of the allele, R_2 , encodes white flowers.

a. Determine the phenotype of each of the following plants:

- R_1R_1

- R_2R_2

- R_1R_2

b. A plant with red flowers is crossed with a heterozygous plant. What proportion of offspring will also have red flowers?

c. Is it possible for the cross described in part b to produce offspring with white flowers? Explain.

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3) Colour blindness is an ~~x~~linked, recessive trait. A colour blind man and a woman homozygous for normal vision have three children: two girls and a boy.

a. Construct a Punnett Square to represent the cross between these two individuals.

b. What is the probability that their son is colour blind?

c. What is the probability that their daughters are both carriers for the allele that encodes colour blindness?

d. What is the probability that they will have colour blind grandchildren?

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- 4) In humans, three different alleles exist for the gene that determines blood type. The allele " I^A " encodes Type A blood, the allele " I^B " encodes Type B blood, and the allele " i " encodes Type O blood. The order of dominance is as follows: $I^A = I^B > i$

a. List all of the possible genotypes for each of the following phenotypes:

- Type A blood =

- Type B blood =

- Type AB blood =

- Type O blood =

b. A woman who is homozygous for blood type A has a child with a man who is heterozygous for blood type B. What is the probability that their child will have blood type AB?